A. Project information

- **Title:** MESAWAVE–Coastally refined global ensemble of ocean wave climate projections forced by storm-permitting CESM simulations
- **Project Leads:** Alper Altuntas, Alice DuVivier, Melissa Moulton (NCAR CGD)
- **Project Collaborators:** Fred Castruccio, Manish Venumuddula (NCAR CGD), Momme Hell (WHOI), Giovanni Seijo-Ellis (UPR)
- **Submission date:** October 9, 2026

B. Project overview

Climate variations on seasonal-to-decadal timescales modulate the likelihood and intensity of extreme weather events including tropical cyclones, winter storms, atmospheric rivers, and floods. In coastal areas, home to 39% of the US population (Sandifer and Scott, 2021), ocean surface waves are an important ingredient in determining the impacts of severe weather events. However, large uncertainties remain in projections of wave climate. Waves travel long distances, communicating weather and climate mode information from one region on the globe to another. Because of this global reach, full global or basin-wide models are often needed to accurately represent the wave climate. Existing global ocean wave projections are limited by the available forcing (e.g., CMIP5, CMIP6 models), which does not adequately resolve storms and mesoscale ocean-atmosphere interactions.

The MESACLIP Project – Understanding the Role of MESoscale Atmosphere-Ocean Interactions in Seasonal-to-Decadal CLImate Prediction – has yielded an unprecedented set of tropical-cyclone-permitting and ocean-mesoscale-eddy-resolving historical and future climate simulations. We propose to use MESACLIP output to force an ensemble of global ocean wave simulations. For a subset of MESACLIP scenarios and time periods, we will apply global unstructured WAVEWATCH III on a 1.67-million node mesh with open-ocean resolution of 25 km and refinement to 10 km on global coastlines and 5 km on US coastlines, costing ~110,000 core-hours per simulated year. Simulations will be conducted for 3 ensemble members across historical (1950-2005), present (2000-2020), mid-century (2040-2060), and end-of-century (2080-2100) conditions (under RCP 2.6, 4.5, 6.0, 8.5) using 6-hr atmosphere and daily ocean–sea-ice MESACLIP outputs. For ensemble member 10, a simulation with available 1-hr atmospheric forcing from 2000-2020 (RCP 8.5) will be run for comparison with observations and the 6-hr-forced results.

This project directly supports the NCAR strategic plan goals “Building National Resilience” and “Understanding the Earth System” by providing a new, unique high-resolution dataset with relevance to wave-driven coastal flood hazards and how those may change in a warming climate. The data from this project will be used to inform community planning and to improve parameterizations in future wave/climate modeling. NCAR is uniquely situated to do this work as a facility with expertise in Earth system modeling across all components, as well as HPC resources needed to generate cutting edge datasets like the proposed MESAWAVE ensemble.

C. Science objectives

The MESACLIP dataset represents an unprecedented collection of global climate simulations featuring tropical cyclone (TC) – permitting (0.25° atmosphere–land resolution) and ocean-mesoscale-eddy–resolving (0.1° ocean–sea-ice) resolution across historical and

future scenarios. The creation of a consistent ocean surface wave dataset with the MESACLIP will bridge critical gaps, addressing the broader objective of connecting atmospheric events, ocean currents, and climate state to localized wave impacts, including coastal flooding, shoreline erosion, and navigation safety. Existing wave climate projections have historically been limited by coarse resolution (Morim et al., 2020; and references therein), restricting their usage to fully address these topics.

Analysis of MESACLIP output has indicated changes in atmospheric rivers, tropical cyclones, mesoscale convective systems, ocean current systems, and regional sea-level patterns (e.g., Shields et al., 2024; Xu et al., 2026). All these phenomena respond to climate change and large-scale climate phenomena like El Niño but also directly perturb surface wave climate. Using wave integrations based on high-resolution, long-term simulations like MESACLIP, we will create an unprecedented dataset to study the individual and compound effects of climate variability across scales on the wave climate and the local hazards we are beginning to observe today. The proposed simulations will allow the scientific community to answer how ocean surface waves respond to these changes and contribute to hazards. Simulating multiple ensemble members across the full range of RCP scenarios, will enable many realizations of weather events and associated wave fields to address this objective. It will also allow to disentangle the forced versus stochastic response in the wave fields. The dataset will also provide consistent wave boundary conditions for dynamical downscaling using the CESM-MOM6-Regional CROCODILE framework. Finally, the dataset is valuable for parameterization development in air-sea interaction research.

There is widespread interest within NCAR and at member universities in using the MESAWAVE product to understand wave climate variability and change, as well as to develop wave-ocean parameterizations and force regional wave-ocean models. At present, three specific initial scientific applications planned with this dataset are: (1) analyzing future changes in global wave climate, including high-resolution information along US coastlines, and its relation to global circulation changes, (2) Understanding how extratropical storm characteristics and their future changes impact the wave climate on the northern coast of the U.S. Caribbean islands, (3) investigating the wave contribution of Arctic storm impacts for changing sea-ice conditions. These projects are in collaboration with coastal communities and NSF NCAR member universities. In project (1), wave climate variability will be analyzed in the context of storm-track shifts and intensity, and ENSO variability. We will link these to characteristic storm size changes. Projects (2) and (3) will contextualize these global changes with local wave condition changes (Ghanavati et al 2024). Extratropical storms, tracked in MESACLIP atmospheric model output, will be clustered based on MESAWAVE spectra at virtual stations along the northern Puerto Rico coast and the Alaskan coast. We plan to identify 'high-impact' storm events, defined partly based on community input, due to waves and other environmental factors, and subsequently quantify future projections of the strength and frequency of these events.

D. Computational experiments & requirements

D1. Computational experiments

D1a. Numerical approach

The modeling framework employs WAVEWATCH III (WW3) (Tolman et al., 2020), the community spectral ocean-surface-wave model used by NOAA/NCEP for operational forecasting and widely adopted within the wave-modeling research community, including

within CESM. WW3 will be run in a standalone, data-forced mode (without interactive ocean or sea-ice) within the CESM/CMEPS coupling infrastructure. WW3 is configured on an unstructured triangular mesh with 25 km open-ocean resolution refined to 5–10 km along the coast (Fig. 1), using the model's PDLIB unstructured-grid capability with an implicit (Jacobi-Gauss-Seidel) solver for spatial propagation. Because the implicit scheme's stability does not depend on local cell size, a single global mesh can span open-ocean and coastal scales without the prohibitively small timestep an explicit scheme would otherwise require; this is what makes the 1.67-million-node mesh computationally tractable at a 600 s timestep (~100,000 core-hours per simulated year, including ocean-current forcing; see Section D1c). This unstructured/implicit configuration follows established prior use of WW3 at comparable global-to-coastal scale (Abdolali et al., 2024, 2026) and within an Earth System Model coupling framework (Brus et al., 2021).

WW3 is forced one-way using pre-computed MESACLIP output fields: 10-m winds from the 0.25° atmosphere component, and daily surface currents and sea-ice concentration and thickness from the 0.1° ocean-sea-ice component. Because MESACLIP does not provide sea-ice floe diameter, which WW3's ice-attenuation source term requires, we derive it diagnostically from concentration and thickness using an empirical fit to CESM3 development output (see Section D1b). From these inputs, WW3 produces the fields this project will archive and publish (Section D1e): integrated wave parameters and the parameters of the three leading wave systems on the global mesh every three hours, and full directional spectra at about 5,000 virtual stations along global coastlines and at historical wave buoy locations. Fig. 2 illustrates this workflow.

D1b. Experiments

The model configuration choices have been selected to adequately resolve open-ocean and regional-scale coastal wave properties.

- Domain: global; Grid size: 25 km with coastal refinement to 5 km; Time step: 600 s
- Key parameter choices:
    - Standard β_{max} wave-growth parameter (Tolman et al., 2020) adjusted for both 6-hr and 1-hr historical MESACLIP forcing to account for extreme wind bias.
    - Sea-ice attenuation is as in CESM3 with parameterized floe diameter (determined via logarithmic fit to CESM3 development data).

An ensemble size of 3 was selected to separate internal and forced variability. RCP scenarios and present, mid-century, and end-of-century time periods were chosen to constrain a range of outcomes relevant for planning. A single ensemble member (10) with available 1-hr atmospheric forcing will be run to assess the impact of the 6-hr forcing on simulated waves including representation of extreme events, and for comparison with historical wave reanalysis outputs that are typically generated with 1–3-hr atmospheric forcing. Ensemble members 5, 9, and 10 were selected based on representation of historical sea-ice metrics and to encompass ensemble spread.

- Historical Experiments:
    - 6-hr forcing: 1950-2005 (3 ensemble members: 5, 9, 10)
    - 1-hr forcing: 2000-2005 (1 ensemble member: 10)
- RCP 8.5 Experiments:
    - 6-hr forcing: 2006-2020, 2040-2060, 2080-2100 (3 ensemble members: 5, 9, 10)
    - 1-hourly forcing: 2006-2020 (1 ensemble member: 10)
- RCP 6.0 Experiments:
    - 6-hr forcing: 2006-2020, 2040-2060, 2080-2100 (3 ensemble members: 5, 9, 10)

- RCP 4.5 Experiments:
    - 6-hr forcing: 2006-2020, 2040-2060, 2080-2100 (3 ensemble members: 5, 9, 10)
- RCP 2.6 Experiments:
    - 6-hr forcing: 2006-2020, 2040-2060, 2080-2100 (3 ensemble members: 5, 9, 10)

D1c. Code performance

We benchmarked the WW3 configuration on Derecho with a 1.67-million-node mesh forced by ERA5 winds. Each benchmark is a 24-day integration with WW3 history output disabled, so it measures the model rather than I/O. Timings come from the CESM/GPTL timing reports written at the end of every job and from the ESMF profiler. We also logged the implicit solver's diagnostics: at 600 s, 79% of timesteps converged within the first 10 iterations and the rest within 20, against a limit of 1000, so no step was close to stalling.

Table 1: Strong scaling of the 1.67M-node test mesh, 600 s timestep, 24-day runs.

Cores	Nodes	s / model day	Sim. years / day	Core-hours / sim. year	Parallel efficiency
2048	16	481.5	0.49	99,982	100%
3072	24	348.1	0.68	108,417	92%
3840	30	309.5	0.76	120,499	83%
4608	36	290.9	0.81	135,905	74%
5376	42	258.3	0.92	140,769	71%

We ran core counts between 2048 and 5376. Throughput rises up to 5376 cores, but so does the cost per simulated year, so we size production jobs in the middle of that range. The timestep was chosen from the same set of 24-day runs: at 5376 cores, a simulated year costs about 224,000 core-hours at 300 s, 140,769 at 600 s (37% less) and 116,896 at 900 s (48% less). Production will use the 1.67-million-node mesh at 600 s on 22 nodes (2816 cores) per job. We budget 110,000 core-hours per simulated year at a throughput of about 0.6 simulated years per day, which includes a small margin over the measured scaling. Although throughput per job is modest, the experiments are independent and will run concurrently: with 8 jobs at a time (176 nodes), all 873 simulated years complete in under six months.

D1e. Output and analysis

The stored data will allow global and detailed regional analysis of the shifting wave climate and the impact of extremes in future decades. The gridded data output follows the established standard of the WW3-based IOWAGA wave hindcast (Ardhuin and Accensi, 2014). Given the widespread interest in the MESAWAVE data within NCAR and from collaborators, we plan to publish globally gridded wave parameters necessary for fundamental wave climate analysis at 3-hourly intervals, including 7 integrated parameters (significant wave height, peak frequency, mean direction, directional spread, mean, zero-crossing, and energy period) and 5 variables representing the 3 leading wave systems. In total, 22 variables require about 1.3 TB per decade. Full spectral data (for 36 frequencies × 24 directions) will be stored at 5000 virtual stations along the global coastlines and historical/current observational stations, requiring about 0.2 TB per decade.

Planned Post-Processing/Analysis steps

CGD OS and PPC Section heads have confirmed that CGD Casper allocations will be available for data analysis and visualization. Global wave partition data will be used to understand shifts in extreme waves and swell under the different climate projections. By differentiating between dominant wind-sea significant wave height and period and their

partitions, we will understand how changing storm-track position, size, and intensity result in extreme wave conditions and hence changing swell heights and propagation angle (Casas-Prat et al. 2024). Further, we will link those wave climate changes to atmospheric circulation patterns that are known to drive storm track variability (Shaw et al. 2016, Meehl et al 2019, Wang et al. 2023).

Virtual station data will be saved at ~5000 global coastal and historical-observation locations to prepare for future studies with partners in different locations. Detailed analysis of MESAWAVE spectra will be performed at virtual stations in locations of interest to our community partners. Historical observed wave spectra will be used to identify spectral characteristics associated with events community members have identified as high impact. We will find spectra matching these characteristics in MESAWAVE spectra time series, yielding a time series of high-impact wave events, including long-period waves known as swells that travel long distances from their source, that can be linked to weather events. Extratropical storms and other weather features that have previously been tracked in MESACLIP atmospheric model output. We plan to assess how these high-impact storm events may be changing in time as a function of sea-ice, ocean state, and other factors.

D2. Resource requirements

D2a. HPC

Table 2. Experiments & estimated core hrs, at 110k core-hrs per sim. year (Section D1c)

Experiment	Forcing	Periods	Members	Sim. yrs	Core-hrs	Stored (TB)
Historical	6-hourly	1950–2005	3	168	18,480k	25.2
Historical	1-hourly	2000–2005	1	6	660k	0.9
RCP8.5	6-hourly	2006–2020, 2040–2060, 2080–2100	3	171	18,810k	25.7
RCP8.5	1-hourly	2006–2020	1	15	1,650k	2.3
RCP6.0	6-hourly	2006–2020, 2040–2060, 2080–2100	3	171	18,810k	25.7
RCP4.5	6-hourly	2006–2020, 2040–2060, 2080–2100	3	171	18,810k	25.7
RCP2.6	6-hourly	2006–2020, 2040–2060, 2080–2100	3	171	18,810k	25.7
Total				**873**	**96M**	**131**

D2b. Campaign Storage

Table 3. Data storage estimates (see Section D1e for justification)

Output	Contents	Raw per decade	Stored per decade	Total stored (TB)
Gridded wave parameters	7 integrated parameters + 3 systems × 5 partition fields (22 variables), 3-hourly	4.1 TB	1.3 TB	113
Station spectra	5,000 stations, 36 frequencies × 24 directions (regridded from the model's 50 × 36), 3-hourly	0.5 TB	0.2 TB	18
Restart files	One per continuous run segment (40×11.5 GB)	–	–	0.4
Total				**131**

Additional supporting information

E. Data management plan

1. Disk Space: Short-term data storage needs include ~80 TB of temporary GLADE scratch space. Scratch holds about two weeks of raw model output (about 0.8 TB per simulated year as written by the model, before repacking, with 8 concurrent jobs) until it is repacked, and additional headroom for test runs and reruns. After completion of the new simulations, scratch space can be reduced back to default. As the integrations are running, output will be post-processed into single-variable, compressed time series and written directly to Campaign Storage. The derived time series used in publications will be added to the archival campaign storage partition to satisfy data transparency requirements.

2. Archival Storage and Data Release:
We anticipate these simulations will be of significant interest to a broad community of researchers with a wide spectrum of interests. CGD OS and PPC leadership have agreed to host the data on their respective partitions within CISL's Campaign Storage system. The dataset will also be made publicly available via Globus Guest Collections, allowing unauthenticated data access to a specified subset of data stored in a mapped collection like NCAR GLADE. In addition, Globus Guest Collections provides basic HTTPS access for downloads directly from a browser or via terminal utilities like wget or curl. The dataset will be promoted through the CESM Ocean Model and Polar Climate Working Groups and documented on the CESM website (http://www.cesm.ucar.edu/projects/community-projects/) alongside other large community projects. Storage and publication on GDEX will also be considered with CGD leadership as a potential next step.

F. Accomplishment report

Prior experience demonstrates the ability of the team to efficiently use large HPC resources on Derecho and manage large datasets:

1. Under the NSF-funded MESACLIP project, which aims to advance our fundamental understanding of the multi-scale dynamical processes underlying S2D predictability, project collaborator Castruccio successfully completed an unprecedented suite of CESM-HR simulations. Through successive large-scale requests, Castruccio has previously been awarded significant computational resources on Derecho, establishing a strong, proven track record of successfully directing large allocations. Beyond executing these simulations, a remarkable achievement of the MESACLIP project has been making the entire dataset publicly available to the community via GDEX. Castruccio led the preparation and curation of these datasets which was a major undertaking.

2. Project co-Lead Altuntas is the software liaison for the CESM Ocean Model Working Group (OMWG) and the lead software developer for the Modular Ocean Model (MOM6) and WAVEWATCH III (WW3) components of CESM. In preparation for this request, he integrated WW3's unstructured-mesh capability into CESM, including support for the

global coastally refined mesh, the mesh-generation tools, and the ERA5 wind forcing used for testing, and carried out the Derecho benchmarks in Section D1c (strong scaling from 2048 to 5376 cores, timestep sensitivity, and output-cost and compression tests).

3. Project co-Lead DuVivier is co-chair of the CESM Polar Climate Working Group (PCWG). As detailed in the 2026 CESM accomplishment report, the PCWG has used development resources for numerous sea-ice-related simulations for CESM3 development. Relevant to this NSC proposal, fully-coupled and ice-ocean model configurations have been used to explore the influence of the new floe-size distribution parameterization within CICE, including coupling to interactive waves, testing wave fracture schemes, and testing wave attenuation methods. PCWG production resources have supported numerous scientific studies to investigate coupled polar system interactions, variability and change in polar regions, and sea-ice processes, as well as cutting-edge sea-ice machine learning parameterization development and assessment.

G. References

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H. Figures and captions

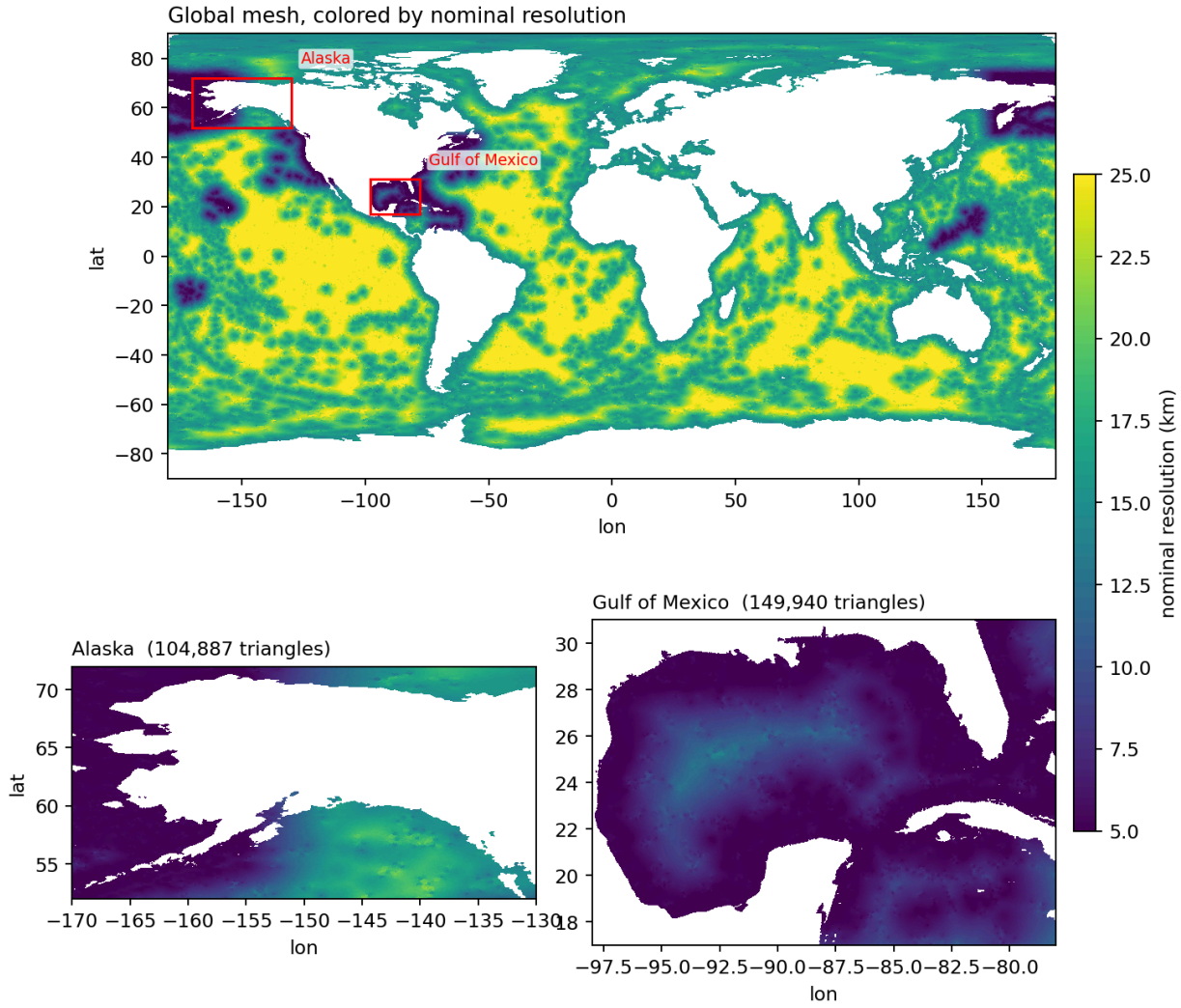


Figure 1. (Top) Global mesh resolution for WW3 unstructured grid, (Bottom) zoom to Alaska and Gulf of Mexico regions.

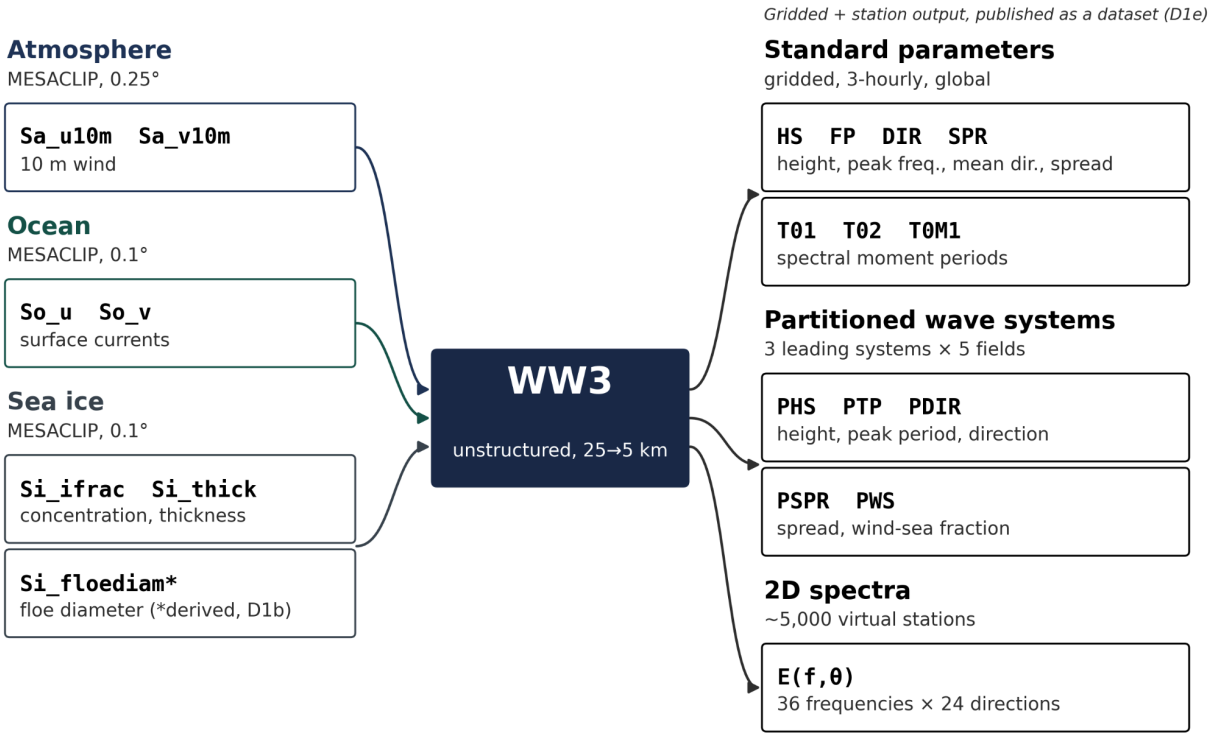


Figure 2. Summary of forcing and outputs for proposed WW3 simulations.